# Siddharth Ray

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## EDUCATION

**University of California, Berkeley** May 2030
*Bachelor of Science in Electrical Engineering and Computer Sciences*
**Vista del Lago High School** May 2026
*High School Diploma, Valedictorian* *GPA: 4.0 UW / 4.43 W*

**Patent:** AI-based DJ system, pending (US 2025/0029584 A1)

## EXPERIENCE

***FIRST* Tech Challenge (FTC) Team #16236 Juice** Folsom, CA
*Vice Captain, Software Division Lead* July 2021 – May 2026

- Developed advanced autonomous, localization, and robot-control systems, including a live relocalization system and sensor-fusion algorithms that achieved <0.1" average positional error over a full match.
- Led technical operations, recruitment, sponsorships, and outreach for a team ranked 9th of 8,000+ worldwide, reaching the 2023 FTC World Championship Division Finals and 2026 World Championship Playoffs; helped raise $8K+/yr and increase rookie retention by 80%.

## PROJECTS

**Automated FTC Virtual Match Reconstruction and Analysis | Computer Vision | GitHub Repo** April 2026

- Developed a lightweight FTC computer-vision pipeline that calibrated match broadcasts to field coordinates and tracked 4 robots through severe occlusion and collisions, enabling automated trajectory reconstruction from standard broadcast footage without GPU-intensive deep learning.
- Built end-to-end analytics tooling for competition scouting, including geometric tracking, merge/identity resolution, shot-event logging, and a local dashboard for calibration and batch processing, reducing manual scouting effort and enabling automated on-location scouting on standard laptops without GPU acceleration.

**Full-Stack Convention/Event Management System | Mobile/Desktop Applications | GitHub Repo** April 2026

- Designed and developed a production event-management platform for a nonprofit, replacing legacy manual workflows with integrated ticketing, POS, meal management, scheduling, maps, and announcements.
- Built cross-platform applications with Flutter (Mobile), Electron.js (Desktop), and a Supabase/PostgreSQL backend, integrating Square payments, Firebase Cloud Messaging, Mapbox, role-based access, and real-time data synchronization.
- Implemented offline-first encrypted local storage for ticketing, check-in, and POS workflows, enabling reliable event operations during network outages and synchronizing transactions when connectivity was restored.

**Ultra-Low Profile Backlit SmartFrame | Embedded Systems | See More** July 2026

- Designed and programmed low-power Arduino firmware with phototransistor-based automatic day/night activation and an RGB status-control interface.
- Engineered an isolated 12V/5V power-distribution system supporting up to 54W of high-density LED lighting within the enclosure's thermal and space constraints.
- Built a 0.75-inch-thick wood enclosure optimizing component layout, heat dissipation, and light diffusion within a compact form factor with an accompanying discrete control panel

**College Basketball Ranking & Game Prediction | PyTorch, pandas, scikit-learn | GitHub Repo** March 2025

- Built an end-to-end ML pipeline that preprocesses NCAA game data, engineers matchup features (travel, rest, home/away, team identity), and trains neural-network models with team embeddings to predict game outcomes.
- Developed a pairwise team-ranking model with time-decayed statistics and calibration/validation metrics (Spearman correlation, log loss, Brier score) to produce rankings and regional win-probability CSV reports.

**DJFlame: AI-Powered Desktop DJ Application | Electron, JavaScript, C++, Firebase | GitHub Repo** February 2022

- Built a cross-platform (MacOS, Linux, Windows) desktop DJ application with Electron.js, featuring real-time guest song requests, YouTube Music queue management, Firebase-backed party synchronization, external-display support, and automatic updates.
- Developed a mobile application using Flutter for cross-platform iOS/Android deployment for guest interactions
- Developed audio-analysis and playback components that estimate track BPM/beat positions and connect to a custom native C++ audio engine using ring buffers, multithreading, track mixing, and bass/treble filtering, laying the groundwork for a planned CNN-driven mixing and automated beat analysis (patent pending).

## SKILLS

**Languages:** Python, C/C++, Java, JavaScript, SQL, Dart, MATLAB

**Frameworks & Software:** OpenCV, PyTorch, Electron, Node.js, Mapbox, Jupyter, Git, PostgreSQL, Flutter, Supabase, Firebase

**Hardware & Embedded:** STM32, ESP32, Arduino, Raspberry Pi, Altium